AP Computer Science Principles Pacing Guide

Forsyth Central High School • 2026–27

Semester 1

Weeks	Unit	Topics / Major Concepts	Projects / Assessments
1-3	Introduction to Computing & Digital Information	Computing systems, hardware/software, binary, data representation, compression, responsible computing	Binary activities, image encoding labs
4-7	Python Fundamentals	Python setup, variables, data types, input/output, operators, debugging	Mad Libs Generator, Tip Calculator, Grade Calculator
8-10	Decision Making	Boolean expressions, if statements, if-else, nested conditionals	Movie Ticket Calculator, Rock Paper Scissors, Quiz Game
11-13	Iteration & Algorithms	for loops, while loops, nested loops, algorithm design, tracing	Number Guessing Game, Pattern Generator
14-16	Lists and Data	Lists, indexing, traversal, aggregation, searching	Gradebook Application, Playlist Manager
17-18	Functions and Modular Programming	Functions, parameters, return values, decomposition	Calculator Functions, Semester 1 Project

Semester 2

Weeks	Unit	Topics / Major Concepts	Projects / Assessments
19-21	Introduction to Pygame	Pygame setup, game loop, events, drawing, coordinates	Interactive Drawing Program
22-24	Game Development Fundamentals	Sprites, animation, keyboard input, collision detection, scoring	Catch Game, Dodge Game
25-27	Create Task Preparation	Program purpose, algorithms, data abstraction, testing	Create Task Practice Project
28-30	Official Create Performance Task	College Board Create Task requirements	Create Task Submission
31-32	Internet & Cybersecurity	Internet architecture, DNS, routing, cybersecurity, encryption	Networking Labs
33-34	Impact of Computing	Ethics, privacy, AI, digital divide, intellectual property	Research Presentation
35-36	AP Exam Review	MCQ review, Create Task written responses, practice exams	Mock AP Exam